

Answer **all** the questions in the spaces provided.

- 1 Ganymede is a moon of mass 1.50×10^{23} kg and radius 2.64×10^6 m orbiting the planet Jupiter of mass 1.90×10^{27} kg as shown in Fig. 4.1. The orbital period of Ganymede is 171.6 hours.

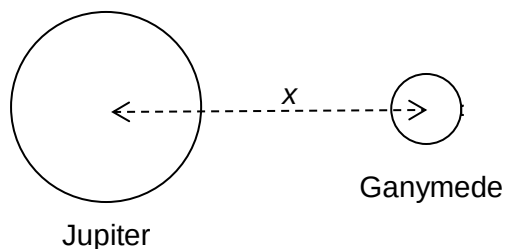


Fig. 4.1

- (a) (i) Show that the distance x between the centres of Jupiter and Ganymede is 1.07×10^9 m. Explain your workings.

Gravitational force by Jupiter on Ganymede provides the centripetal force on Ganymede to orbit Jupiter

$$\frac{GMm}{x^2} = mx \left(\frac{2\pi}{T} \right)^2$$

$$\frac{GMT^2}{4\pi^2} = x^3$$

$$x = \sqrt[3]{\frac{(6.67 \times 10^{-11})(1.90 \times 10^{27})(171.6 \times 3600)^2}{4\pi^2}}$$

$$x = 1.07 \times 10^9 \text{ m}$$

[2]

- (ii) S is a point between Jupiter and Ganymede where the resultant gravitational field strength is zero. Determine the distance from the centre of Jupiter to point S.

$$g_G - g_J = 0$$

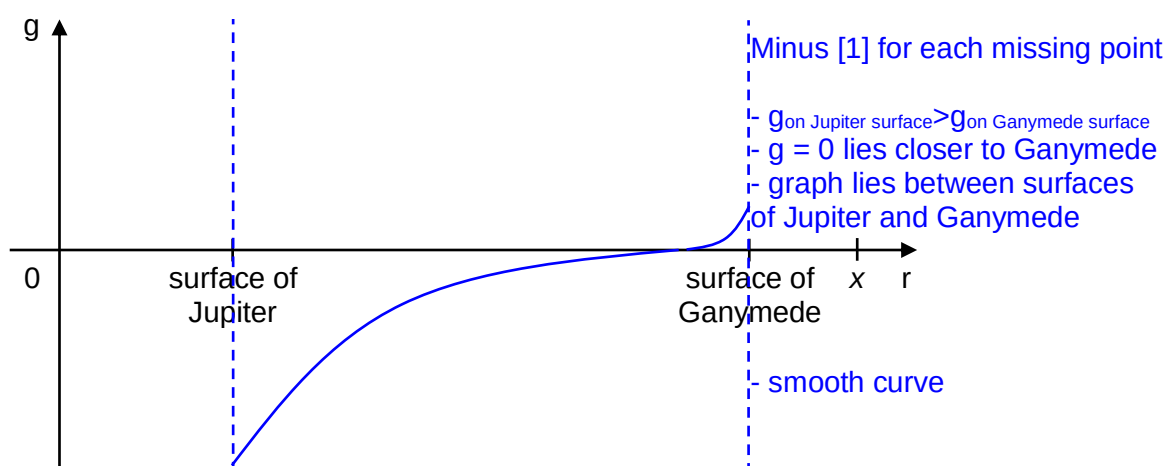
$$\frac{Gm}{(x-s)^2} = \frac{GM}{s^2}$$

$$\frac{s}{1.07 \times 10^9 - s} = \sqrt{\frac{1.90 \times 10^{27}}{1.50 \times 10^{23}}}$$

$$s = 1.06 \times 10^9 \text{ m}$$

$s = \dots\dots\dots \text{ m}$ [2]

- (iii) On the given axes below, sketch a graph from the surface of Jupiter to the surface of Ganymede to show the variation of gravitational field strength g with distance r from the centre of Jupiter. [2]



- (b) A rock is positioned on the surface of Ganymede at the point furthest from Jupiter, as shown in Fig. 4.2. The rock is launched with an escape velocity v for it to reach infinity, far away from Jupiter and Ganymede.

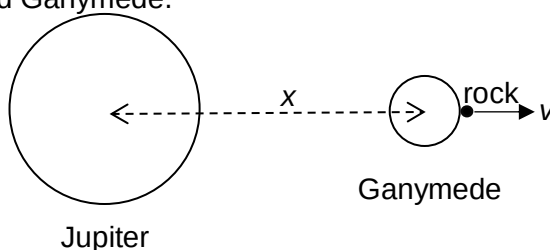


Fig. 4.2

- (i) Show that the gravitational potential at the position of the rock in Fig. 4.2 is $-1.22 \times 10^8 \text{ J kg}^{-1}$.

$$\begin{aligned}\phi &= -\frac{GM}{x+r_G} + -\frac{Gm}{r_G} \\ &= -\left(6.67 \times 10^{-11}\right) \left(\frac{1.90 \times 10^{27}}{1.07 \times 10^9 + 2.64 \times 10^6} + \frac{1.50 \times 10^{23}}{2.64 \times 10^6} \right) \\ &= -1.22 \times 10^8 \text{ J kg}^{-1}\end{aligned}$$

[2]

- (ii) Calculate the escape velocity v .
loss in KE = gain in GPE

$$\begin{aligned}\frac{1}{2} m_r v^2 - 0 &= 0 - m_r (-1.22 \times 10^8) \\ v &= 1.56 \times 10^4 \text{ m s}^{-1}\end{aligned}$$

$v = \dots\dots\dots \text{ m s}^{-1}$ [2]

2 A metal block of 15 cm height was partially submerged in water of density $1.0 \times 10^3 \text{ kg}$